



Distribution parameter alignment-based health indicator for degradation assessment and incipient fault detection of wind turbine bearings under non-stationary operating conditions

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ABSTRACT

Most existing health indicator (HI) construction models are typically developed based on the assumption of constant operating conditions. However, such models are often insufficient for accurately characterizing the dynamic deterioration process and are not conducive to incipient fault detection (IFD) under non-stationary working conditions. To address the above limitations, this paper presents a new distribution parameter alignment-based health indicator (DPA-HI) for degradation assessment and IFD of wind turbine (WT) bearings. Specifically, a parameter alignment function (PAF) is introduced to transform the parameters from non-reference conditions to a reference condition. Subsequently, the component parameters of both the Gaussian and the exponential mixture models under non-reference speeds are aligned to the reference speed using the designed PAF. Based on the aligned parameters, time-domain and frequency-domain HIs are independently derived, and the final DPA-HI is obtained by linearly combining these two HIs. Finally, Experimental results obtained from WT bearing datasets indicate that the DPA-HI effectively captures the intrinsic degradation process under non-stationary conditions. Furthermore, it outperforms both traditional and advanced HIs, facilitating accurate detection of incipient faults in WT bearings.

1. Introduction

With the increasing global demand for sustainable energy, wind power has merged as a key component of the clean energy portfolio. However, owing to the harsh operating environment, wind turbine (WT) bearings are highly prone to failure, resulting in substantial operation and maintenance costs [1–3]. Consequently, degradation assessment and incipient fault detection (IFD) of WT bearings are essential for ensuring their reliability and efficiency. Among various approaches, health indicator (HI) construction serves as a crucial step, providing a quantitative representation of performance degradation and facilitating IFD of WT bearings [4,5]. A high-quality HI not only effectively characterizes the degradation trajectory, but also accurately detect the IFD for health status division [6–8].

In the past decade, numerous HI construction methodologies have been investigated to support the health condition monitoring of machinery. From the perspective of technical level, existing HI construction

approaches can generally be categorized into four categories: statistical parameter-based methods, signal preprocessing-based methods, traditional machine learning-based methods, and deep learning-based methods [5]. Statistical parameter-based methods primarily utilize statistical parameters of monitoring signals as HIs to reflect the health status of mechanical components. Commonly adopted parameters primarily include traditional statistical features, sparsity measures and complexity indicators. Among traditional statistical features, root mean square (RMS), kurtosis and skewness have been widely utilized to evaluate the nonlinear degradation process of mechanical components [9–11]. For instance, Li et al. [12] applied 3σ principle and kurtosis to adaptively identify the first prediction point, while RMS was employed to characterize bearing degradation trends. Similarly, Huang et al. [13] also adopted RMS to capture the deterioration trajectories of bearings. As for sparsity and complexity parameters, the Gini index (GI) [14], negative entropy (NE) [15], L2/L1 norm [16], and Hoyer measure (HM) [17] have been introduced as HIs to describe the deterioration process of

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machinery. The statistical parameter-based HIs are generally statistical features related to mechanical fault characteristics, which can reflect the evolution of performance degradation to a certain extent, but they are susceptible to complex noise interference, resulting in inaccurate representation of the degradation process of mechanical components.

To accurately characterize mechanical degradation and improve the extraction of fault-sensitive information from monitoring signals, signal preprocessing-based methods have been developed to eliminate irrelevant components. The commonly used preprocessing approaches include litering, differencing, Fourier transform, envelope analysis. For example, Zhang et al. [18] introduced a degradation indicator using kurtosis and band-pass filtered data, which was then utilized to estimate the life of machinery. Similarly, Hu et al. [19] developed a HI derived from band-pass filtering and envelope analysis to evaluate the impact of wear on gear performance. By leveraging signal processing techniques, signal preprocessing-based methods can effectively remove irrelevant components from monitoring data, thereby enabling the extraction of relevant statistical features to characterize the fault evolution process of machinery. Nevertheless, the data scales of different features are significantly different, and single statistic feature has the problems of incomplete information representation and uneven contribution for HI construction.

Conventional machine learning-based methods have been explored to address the limitations of signal preprocessing-based approaches. Among them, data dimensionality reduction algorithms have been widely employed to construct HIs for health condition monitoring of machinery. For example, principal component analysis (PCA) was utilized by Widodo et al. [20] to reduce the dimensionality of extracted features, and bearing HIs were subsequently derived by evaluating the distance deviation between reduced features from healthy and unknown states. Pan et al. [21] developed a HI for gearbox health monitoring by fusing multi-sensor data through kernel principal component analysis (KPCA). Qin et al. [22] constructed a HI using isometric mapping (ISOMAP) to assess the degradation of gear. Although these methods can integrate multiple statistical features to provide a more holistic depiction of machinery degradation, they generally require special expert knowledge to select effective features before linear or nonlinear dimensional reduction of features. In addition, inappropriate statistical features will hinder the construction of fusion HIs.

With significant advances in computational power and algorithmic design, deep learning-based methods have emerged as an effective means to address the constraints of traditional machine learning in HI construction. From the perspective of training strategies, deep learning-based approaches can be generally divided into supervised and unsupervised methods. In supervised approaches, specific HI labels need to be predefined for model training, enabling the trained model to extract degradation-representative information and output meaningful HIs. For example, Guo et al. [23] presented a deep convolutional neural networks (DCNNs)-based method to mitigate the trend burr problem in HIs. Chen et al. [24] developed a trainable CNN to construct HIs that describe bearing degradation over time. Cheng et al. [25] introduced a CNN-based approach for health evaluation, in which a nonlinear degradation index was utilized as the training label. This approach enabled the network to establish a mapping between vibration signals and corresponding degradation indicators, thereby enhancing the accuracy of performance assessment. As for unsupervised approaches, HI labels and prior knowledge are not required during the process of HI construction. Therefore, these methods have an inherent advantage over the supervised methods, and can generate HIs from monitoring data without supervision. For instance, Chen et al. [26] presented an embedded convolutional auto-encoder network to mine trend features from monitoring data, and the extracted features were then fused to obtain bearing HIs. Mao et al. [27] designed a deep tensor auto-encoder-based HI for degradation state assessment of metro wheel. Li et al. [28] developed a shape-constrained neural data fusion network, producing HIs with strong trend consistency. Guo et al. [29] proposed an

unsupervised temporal convolutional network based auto-encoder for generating HI from vibration signals. Nevertheless, unsupervised methods may be limited when full life-cycle data are scarce, potentially reducing their effectiveness in constructing reliable HIs.

The Gaussian mixture model (GMM) has been applied for HI construction by characterizing data distributions, enabling effective estimation of distribution parameters from scarce monitoring data across different health status. In recent years, several researches about GMM-based HI construction have been investigated. For example, Wang et al. [30] constructed a HI combining feature selection with GMM. In this method, the GMM was employed to estimate the characteristic distributions under normal and monitored conditions respectively, and then the confidence value was utilized to measure the overlap area between distributions. Rai et al. [31] introduced a HI via GMM and Jensen-Rényi divergence (JRD), where GMM was adopted for data distribution estimation, and the JRD was used to measure the difference between distributions to establish bearing HIs. Wen et al. [32] developed a GMM-based HI construction method for early fault detection of bearing, estimating vibration signal distributions and using Wasserstein distance to quantify deviations between initial and current health states. Zhu et al. [33] estimated the data distributions using GMM, and then constructed HI by minimizing the distribution transport cost. Li et al. [34] constructed an unsupervised bearing HI by establishing a feature model combining GMM and fuzzy entropy. Although these methods can effectively construct HIs under constant or near-constant operating conditions, some mechanical equipment in practical engineering operate under variable working conditions. For example, the working conditions of WT fluctuate with wind speed, leading to unstable speed and load. Such time-varying speed conditions (TVSCs) introduce amplitude and frequency modulation into monitoring data, which will compromise the accuracy of HI in reflecting machinery degradation. Therefore, it is essential to construct HI to characterize the degradation evolution of machinery under TVSCs.

In recent years, some researches on eliminating the effect of TVSCs for health condition monitoring have been explored. For example, Sim et al. [35] utilized isolation forest (IF) algorithm to generate HI under highly variable working conditions. However, the constructed HI has poor degradation assessment capability owing to the inherent randomness of the IF approach. Zhou et al. [36] developed a conditional feature decoupling learning framework to mitigate the anomaly detection interference caused by the coupling between operating conditions and health statuses. Rao et al. [37] proposed a speed normalization auto-encoder model, in which a speed normalization branch can normalize the initial monitoring data to eliminate the impact of speed variations. Li et al. [38] established a baseline transformation algorithm to convert the observed RMS values under different rotational speeds to the baseline speed, successfully mitigating the influence of rotational speed on observed data. Zhang et al. [39] developed an amplitude normalization strategy to suppress rotational speed interference, and HI was constructed from distribution perspective. Sun et al. [40] proposed a dual-decoupling network to effectively mitigate the interference of working conditions. It explicitly mapped the relationship between the working conditions and normal responses, and then deeply decoupled condition features from anomaly features using an adversarial loss. Although these methods have achieved notable progress in alleviating the effect of TVSCs on HI construction, they generally fail to reduce the interference of TVSCs from the perspective of distribution parameters and remain constrained by the requirement for full life-cycle data. To address the aforementioned limitations, a novel HI construction framework based on distribution parameter alignment is proposed to characterize the degradation trajectories of WT bearings under TVSCs. Considering that mechanical load is typically associated with rotational speed, this work primarily focuses on mitigating the influence of speed variations on distribution parameters. First, the parameter alignment function (PAF) is introduced to convert the distribution parameters at non-reference speeds to a designated reference speed. Subsequently, the

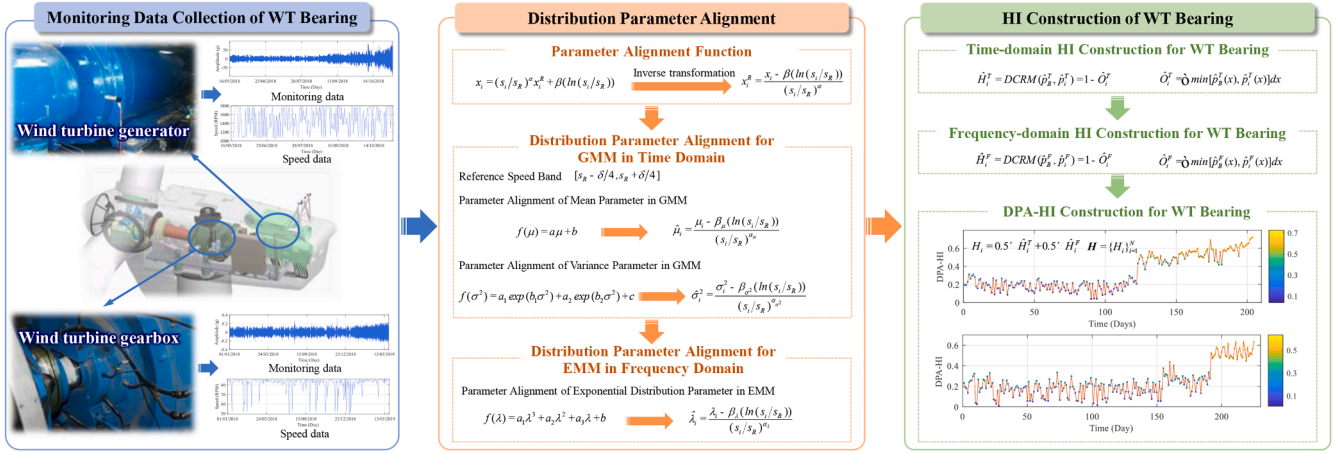


Fig. 1. The overall framework of the developed method.

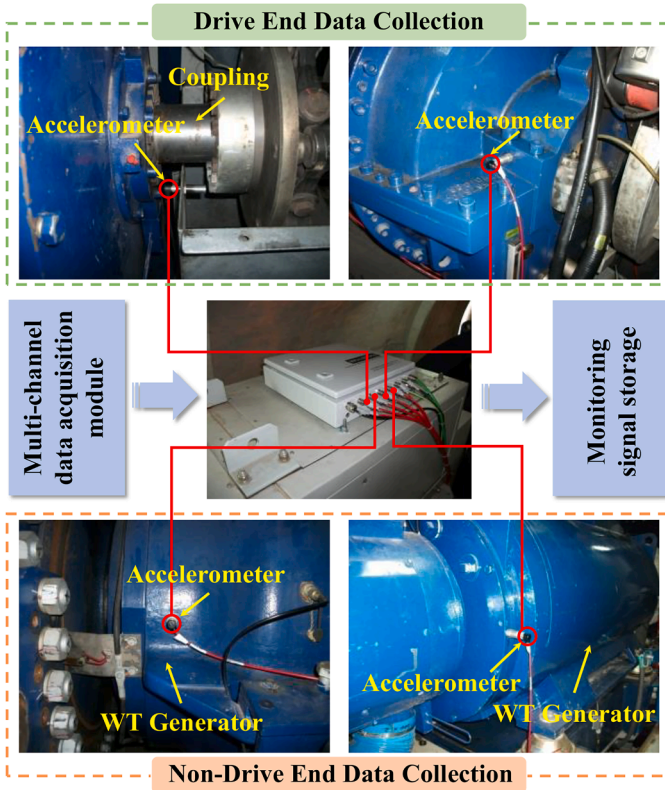


Fig. 2. The sensor configuration for data acquisition.

component parameters of GMM and exponential mixture model (EMM) estimated under non-reference speeds are transformed to the reference speed using the proposed PAF, respectively. Based on the aligned parameters, time-domain HI (THI) and frequency-domain HI (FHI) are independently acquired by the calculated distribution discrepancies. These two HIs are then fused to generate the distribution parameter alignment-based HI (DPA-HI). The developed DPA-HI effectively captures the intrinsic degradation pattern under variable speeds and enables accurate IFD of WT bearing. Finally, the performance and advantage of the DPA-HI are validated using two WT bearing datasets. The main contributions of this work are listed as follows:

(1) A novel PAF is designed to achieve a unified representation of distribution parameters under variable speed conditions. Using the developed PAF, distribution parameters estimated within a speed

interval can be consistently aligned to the reference speed.

(2) The distribution parameter alignment-based HI construction method is proposed to characterize the degradation behavior of WT bearings under TVSCs. Specifically, the component parameters of GMM and EMM are aligned using the designed PAF, based on which THI and FHI are independently constructed from the aligned model parameters. These two HIs are subsequently linearly fused to generate the proposed DPA-HI.

(3) The degradation assessment performance of the proposed DPA-HI under TVSCs are validated using two WT bearing datasets. The developed HI can quantitatively assess the degradation process and accurately detect the IFD of WT bearing, demonstrating strong capabilities in both degradation assessment and fault detection.

2. Distribution parameter alignment-based HI construction method

In this section, a PAF is first designed to achieve DPA under TVSCs. Subsequently, the component parameters of GMM and EMM at non-reference conditions are aligned to the reference condition using the designed PAF. Based on the aligned parameters, THI and FHI are constructed, respectively. These two HIs are then linearly combined to form the proposed DPA-HI, which can effectively capture degradation trends and enable accurate IFD of WT bearings. The overall framework of the developed method is illustrated in Fig. 1.

2.1. Design of parameter alignment function

In mechanical health condition monitoring, mixture models have been widely adopted to estimate the probability distributions of monitoring data for HI construction. However, due to the variations in wind speed within wind farm, the collected monitoring data of WT bearing will be disturbed by speed fluctuation. Such interference may adversely affect the accuracy of mixture models in estimating the true data distribution. To mitigate the effect of speed fluctuations on distribution parameters, the straightforward method is to define a reference operating condition and transform distribution parameters obtained under non-reference conditions to this reference state through a parameter alignment process. Therefore, the design of an effective PAF is crucial for achieving a unified representation of distribution parameters across time-varying conditions. The distribution parameters are assumed as $\{x_i\}_{i=1}^N$, the corresponding time instants and rotating speed records are $\{t_i\}_{i=1}^N$ and $\{s_i\}_{i=1}^N$. It is generally considered that the distribution parameter x_i under speed s_i is calculated by the scale and translation operations of the reference distribution parameter x_k^R . The generalized formula expression is written as follows:

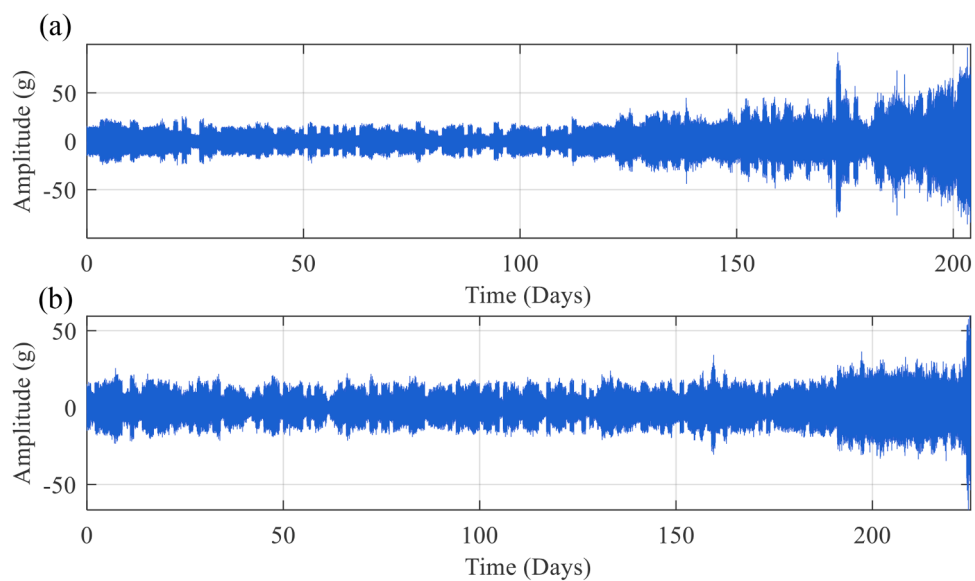


Fig. 3. The corresponding time-domain vibration signals. (a) Generator 1, (b) Generator 2.

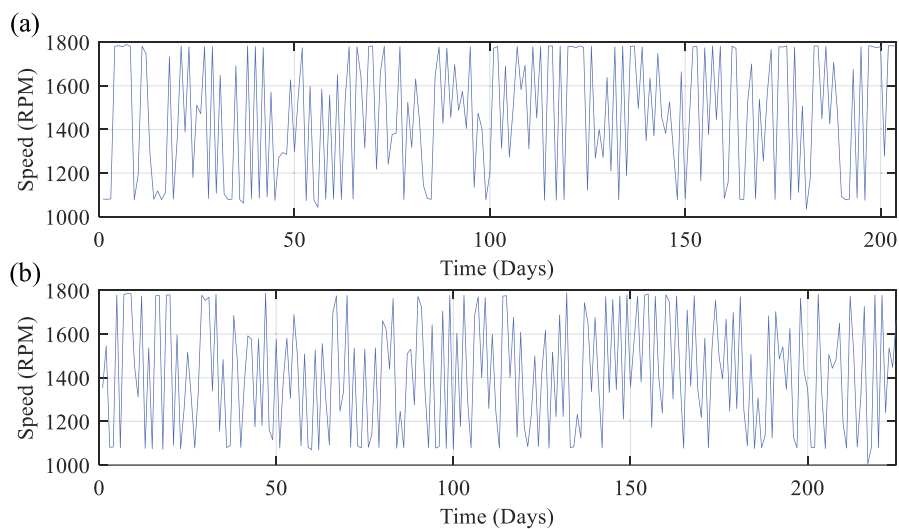


Fig. 4. The corresponding rotational speed profiles. (a) Generator 1, (b) Generator 2.

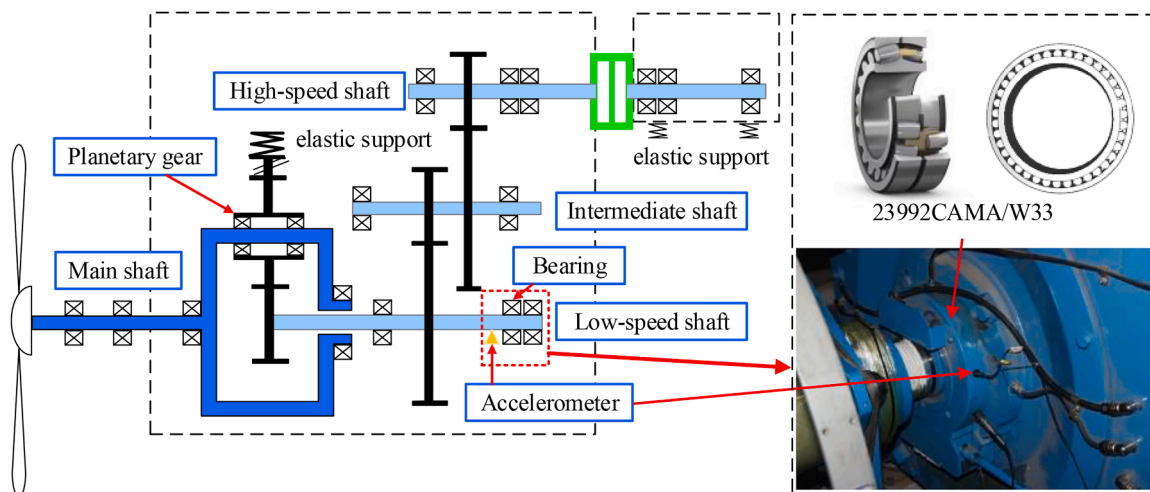


Fig. 5. The simplified schematic of WT gearbox.

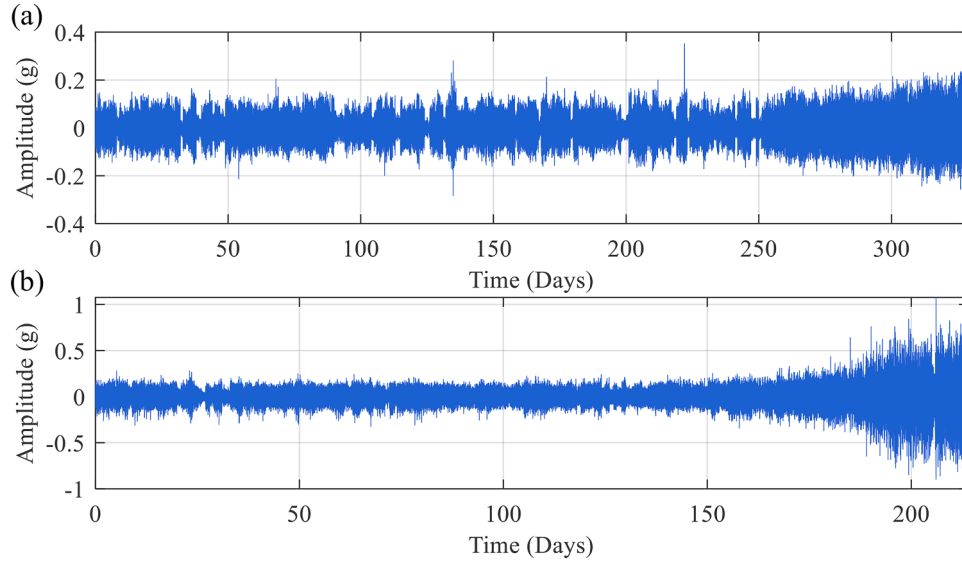


Fig. 6. The corresponding time-domain vibration signals. (a) Gearbox 1, (b) Gearbox 2.

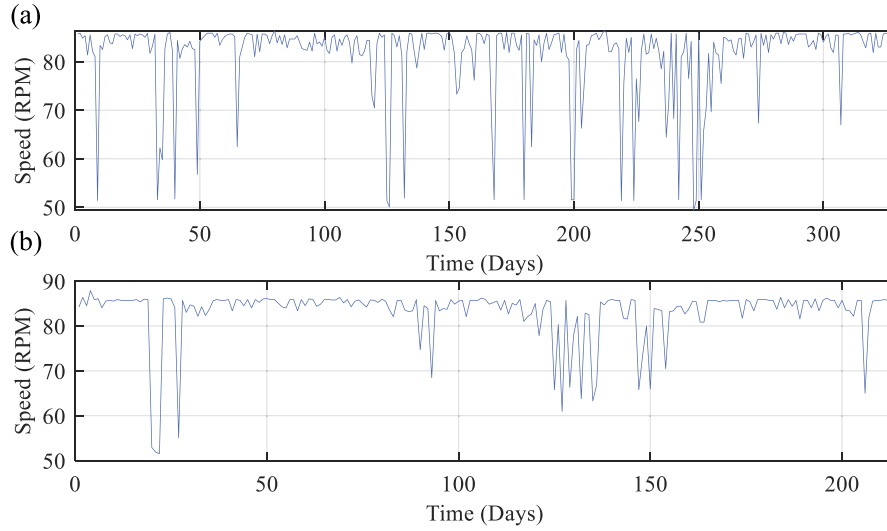


Fig. 7. The associated rotational speed profiles. (a) Gearbox 1, (b) Gearbox 2.

$$x_i = \omega(s_i)x_i^R + \xi(s_i), \quad (1)$$

where $\omega(\cdot)$ denotes the scale transformation factor and $\xi(\cdot)$ represents the translation transformation factor, both of which are continuous functions of the rotational speed s_i . When s_i equals the predefined reference speed s_R , the distribution parameter x_i should coincide with the reference distribution parameter x_i^R . Accordingly, the scale and translation transformation factors must satisfy the following constraints: $\omega(s_R) = 1$ and $\xi(s_R) = 0$. Under these constraints, a commonly used power-law form $\omega(s_i) = (s_i/s_R)^\alpha$ is adopted as the scale transformation factor, while a logarithmic form $\xi(s_i) = \beta(\ln(s_i/s_R))$ is selected as the translation transformation factor. Consequently, Eq. (1) can be rewritten as follows:

$$x_i = (s_i/s_R)^\alpha x_i^R + \beta(\ln(s_i/s_R)), \quad (2)$$

where α and β are two adjustable parameters that enhance the flexibility of the aforementioned function. Consequently, the distribution parameters estimated at arbitrary rotational speeds can be mapped to the reference speed through the following PAF:

$$x_i^R = \frac{x_i - \beta(\ln(s_i/s_R))}{(s_i/s_R)^\alpha}, \quad (3)$$

where x_i^R denotes the reference distribution parameter at time t_i .

In general, the reference distribution parameter at the reference speed can be directly acquired using mixture models during the monitoring process. However, the reference distribution parameters corresponding to non-reference speeds cannot be directly obtained by mixture models and are regarded as virtual values. These virtual reference parameters can only be approximated through interpolation or fitting based on the known reference distribution parameters. Denoting the virtual reference distribution parameters at non-reference speeds as x'_j , the estimation of PAF parameters is achieved through the solution of the following optimization problem:

$$\min_{\alpha, \beta} \sum_{j=1}^M \left(\frac{x_j - \beta(\ln(s_j/s_R))}{(s_j/s_R)^\alpha} - x'_j \right)^2, \quad (4)$$

where x_j and s_j denote the distribution parameter and the corresponding rotational speed at time t_j , respectively. x'_j denotes the virtual reference

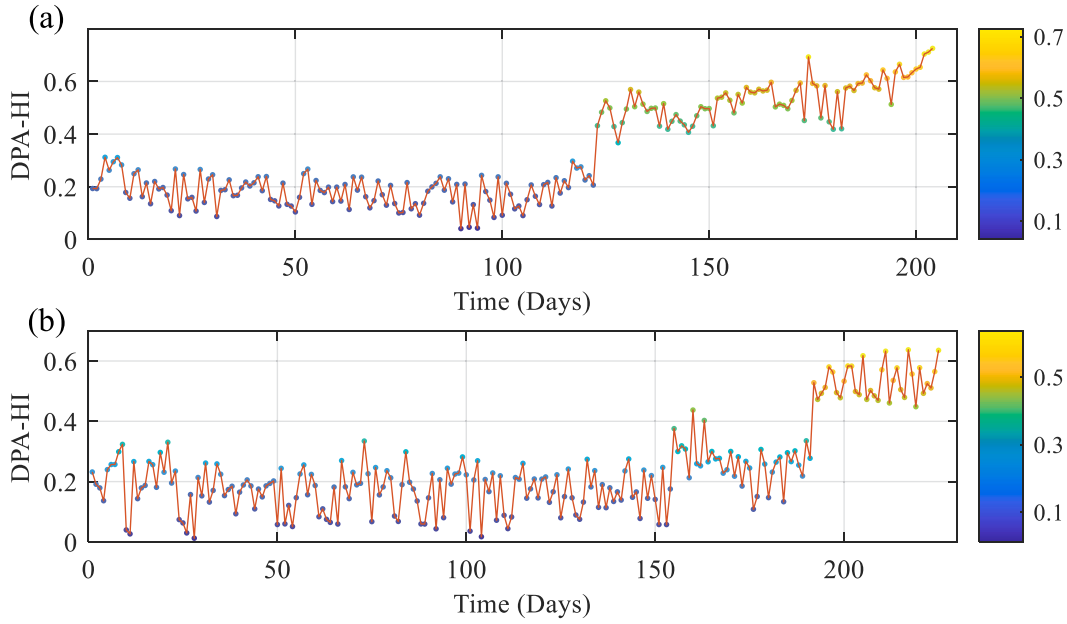


Fig. 8. The acquired generator bearing HIs. (a) Generator 1, (b) Generator 2.

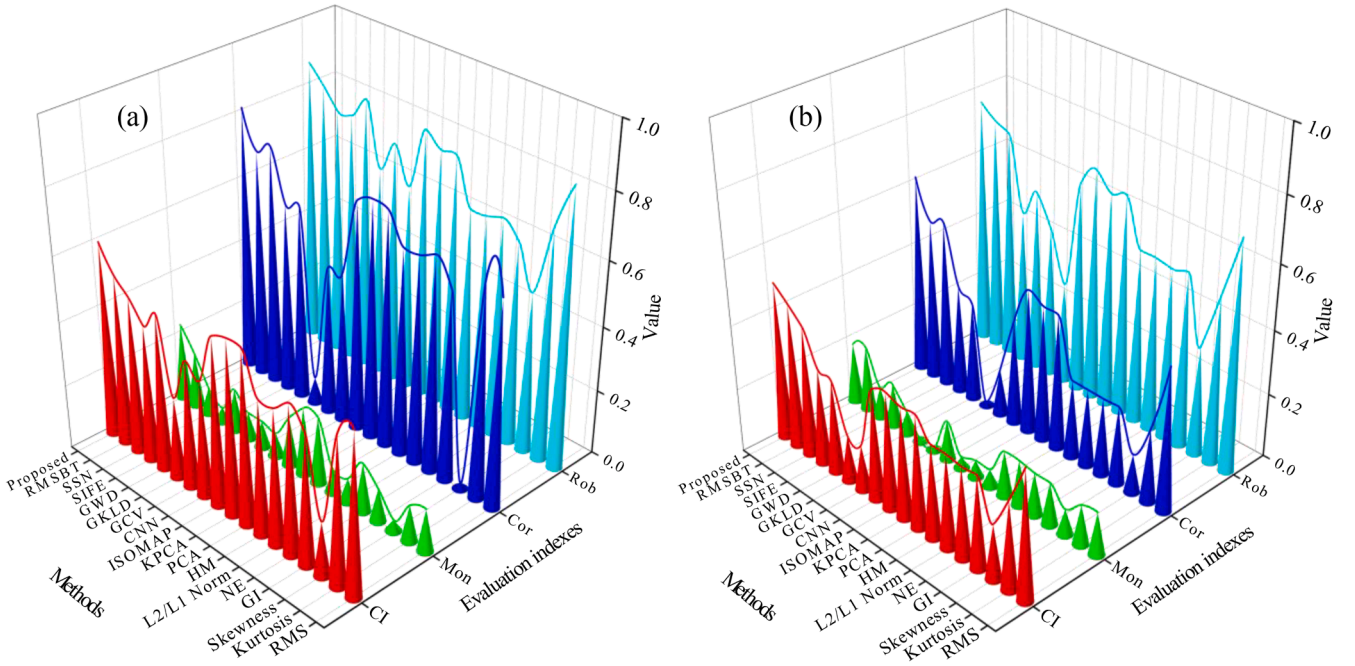


Fig. 9. The evaluation results for WT generator bearing datasets. (a) Generator 1, (b) Generator 2.

distribution parameter at time t_j under the reference speed, M is the number of distribution parameters under non-reference speeds. The objective function can be solved using the nonlinear least-squares optimization technique, which ensures rapid convergence and high estimation accuracy.

2.2. Distribution parameter alignment based on PAF

In this section, the distribution parameters of GMM in the time domain are first aligned. Specifically, the mean and variance parameters are aligned using the PAF. Subsequently, the distribution parameter of EMM in the frequency domain is aligned, where the exponential distribution parameter is also aligned via the PAF.

2.2.1. Distribution parameter alignment for GMM in time domain

For HI construction, the GMM is utilized to characterize the data distribution by estimating and extracting the relevant distribution parameters from monitoring data. Among all mixture components, the one with the maximum mixture weight is primarily selected to characterize the data distribution of WT bearing. The dominant component's parameters are defined by the mean and variance of the corresponding distribution. The monitoring data of WT bearing is denoted as $\mathbf{y} = \{\mathbf{y}_i\}_{i=1}^N$, $\mathbf{y}_i = \{\mathbf{y}_k\}_{k=1}^n$, where N represents the total number of samples and n denotes the data length of each collected sample. The corresponding speed records are denoted as $\mathbf{s} = \{\mathbf{s}_i\}_{i=1}^N$. Based on the WT bearing monitoring data, the GMM's parameters are iteratively optimized via the expectation maximization algorithm. The mean and

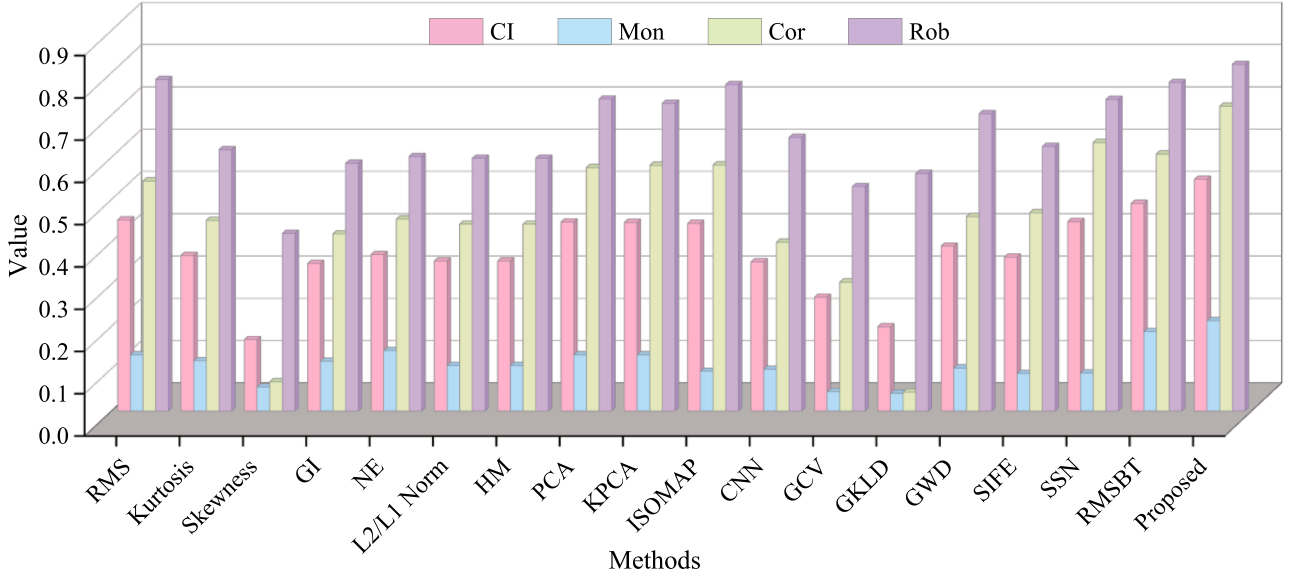


Fig. 10. The average evaluation results for WT generator bearing datasets.

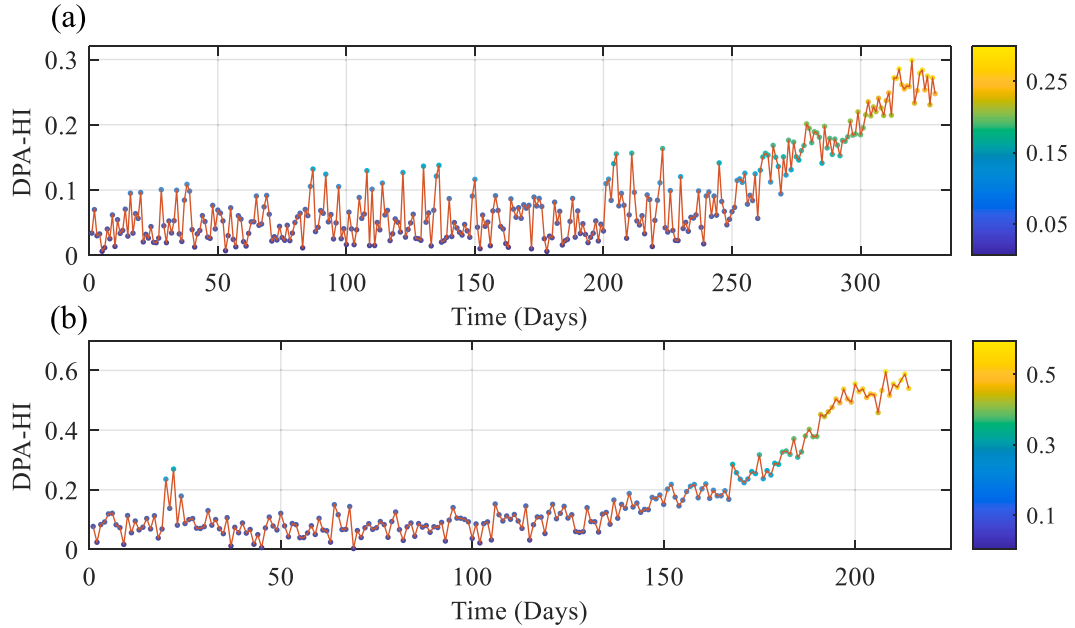


Fig. 11. The constructed gearbox bearing HIs. (a) Gearbox 1, (b) Gearbox 2.

variance of GMM are then obtained as $\mu = \{\mu_i\}_{i=1}^N$ and $\sigma^2 = \{\sigma_i^2\}_{i=1}^N$, respectively.

To perform distribution parameter alignment for GMM in time domain, a reference speed must first be specified. In principle, the reference speed can be arbitrarily chosen within the span of the recorded speeds. In this work, the reference speed s_R is determined as the mean value of all recorded speeds $s = \{s_i\}_{i=1}^N$. To account for transmission uncertainties and measurement errors, a reference speed band centered around s_R is further defined as follows:

$$[s_R - \delta/4, s_R + \delta/4], \quad (5)$$

where s_R denotes the reference speed, and δ denotes the interquartile range of the recorded speeds. This parameter characterizes the dispersion of the central 50% of the recorded speeds and is robust to outliers. The bandwidth is defined as $\delta/4$, centered around the reference speed.

Using $\delta/4$ defines a relatively narrow band around the reference speed, which effectively captures the most stable operating condition while excluding transient fluctuations and extreme values. Through extensive experiments, we observed that using $\delta/4$ as the bandwidth yields more stable and effective HIs compared to wider bandwidths, especially under non-stationary conditions. Using the defined reference speed band, the reference mean values and non-reference mean values can be identified. To obtain the reference mean values corresponding to non-reference speeds, a linear function is defined as follows:

$$f(\mu) = a\mu + b. \quad (6)$$

The parameters a and b are initially estimated using the reference mean values. After that, the PAF parameters for the mean of Gaussian distribution are estimated via the mean values, reference mean values, and speed records at the non-reference speeds. Finally, the mean parameters after parameter alignment $\hat{\mu} = \{\hat{\mu}_i\}_{i=1}^N$ are obtained through the

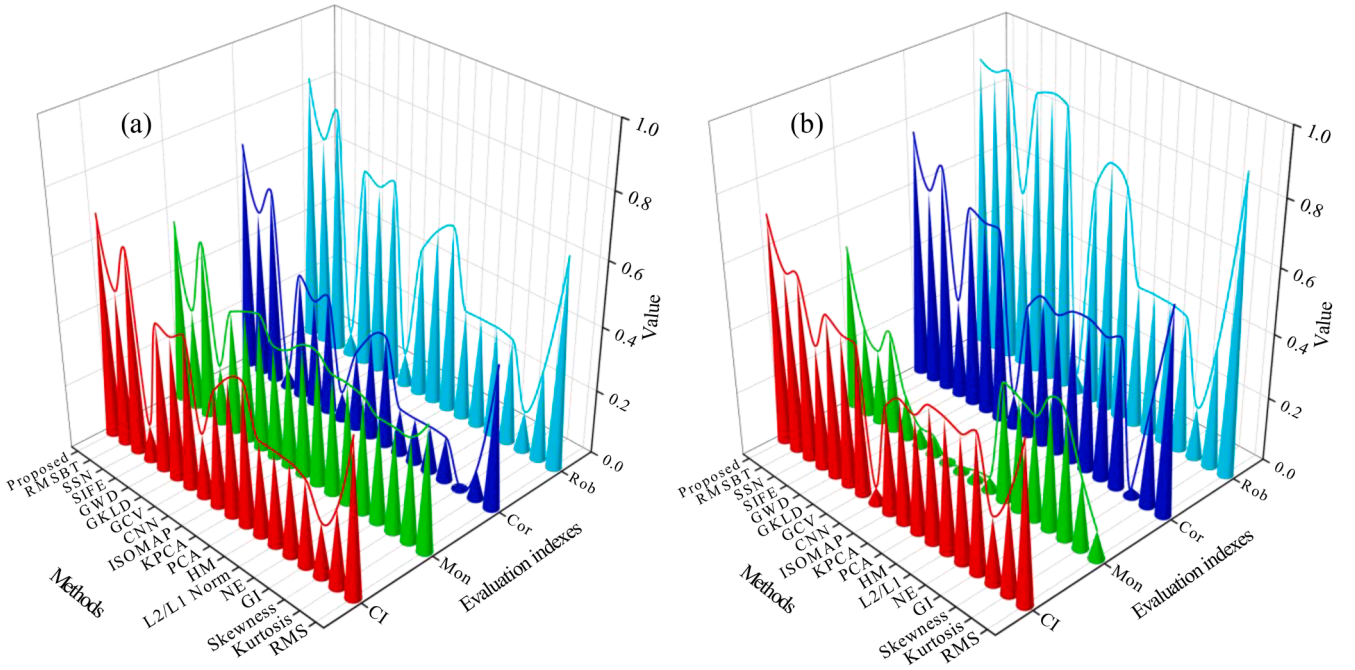


Fig. 12. The evaluation results for WT gearbox bearing datasets. (a) Gearbox 1, (b) Gearbox 2.

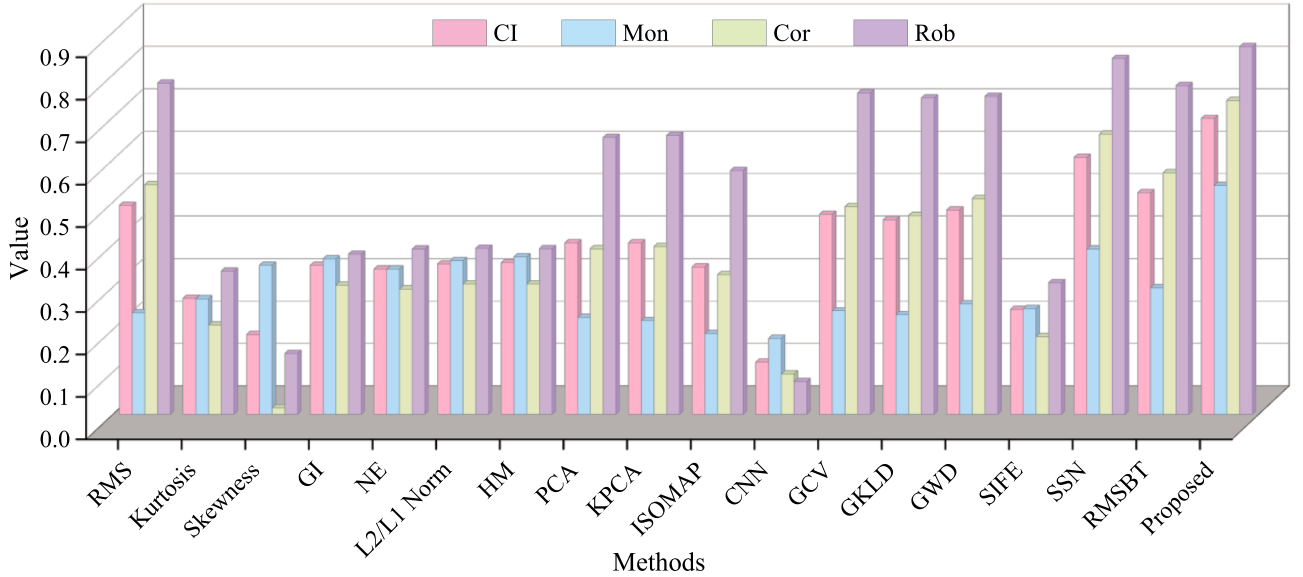


Fig. 13. The average evaluation results for WT gearbox bearing datasets.

fitted PAF as follows:

$$\hat{\mu}_i = \frac{\mu_i - \beta_\mu (\ln(s_i/s_R))}{(s_i/s_R)^{\alpha_\mu}}, \quad (7)$$

where $\hat{\mu}_i$ denotes the mean parameter after parameter alignment, α_μ and β_μ represent the estimated PAF parameters corresponding to the mean parameter of Gaussian distribution.

In addition to the mean parameter, the variance parameter also requires parameter alignment. The reference and non-reference variance values can be identified according to Eq. (5). The reference variance values are then used to fit a double-exponential model to estimate the reference variances at non-reference speeds. The double-exponential model is defined as follows:

$$f(\sigma^2) = a_1 \exp(b_1 \sigma^2) + a_2 \exp(b_2 \sigma^2) + c, \quad (8)$$

where a_1 , b_1 , a_2 , b_2 , and c are the model parameters. Based on the reference variances at non-reference speeds, the parameters of PAF for the variance parameter of Gaussian distribution are estimated using the variance values and speed records at non-reference speeds. Finally, the variance parameters after parameter alignment $\hat{\sigma}^2 = \{\hat{\sigma}_i^2\}_{i=1}^N$ are acquired through the fitted PAF as follows:

$$\hat{\sigma}_i^2 = \frac{\sigma_i^2 - \beta_{\sigma^2} (\ln(s_i/s_R))}{(s_i/s_R)^{\alpha_{\sigma^2}}}, \quad (9)$$

where $\hat{\sigma}_i^2$ is the variance parameter after parameter alignment, α_{σ^2} and β_{σ^2} are the estimated parameters of PAF corresponding to the variance

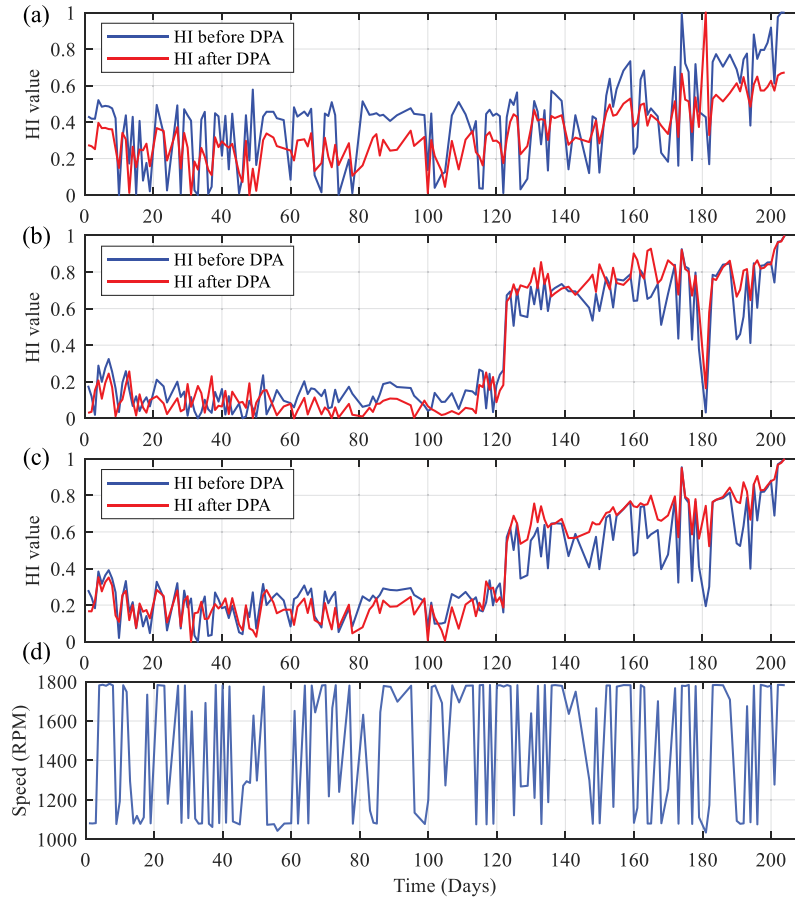


Fig. 14. The constructed HIs and rotational speed profiles under non-reference speeds. (a) THI, (b) FHI, (c) Final HI, (d) Rotational speed profiles used for alignment.

Table 1

The evaluation results and confidence interval of different methods for all datasets.

Method	Mon	Confidence Interval	Cor	Confidence Interval	Rob	Confidence Interval	CI	Confidence Interval
RMS	0.186	[0.111 0.326]	0.543	[0.445 0.639]	0.782	[0.684 0.880]	0.472	[0.423 0.500]
Kurtosis	0.195	[0.119 0.273]	0.330	[0.161 0.585]	0.478	[0.337 0.666]	0.321	[0.266 0.421]
Skewness	0.204	[0.057 0.366]	0.042	[0.013 0.098]	0.282	[0.143 0.420]	0.179	[0.169 0.200]
GI	0.242	[0.118 0.387]	0.362	[0.192 0.531]	0.482	[0.377 0.586]	0.350	[0.280 0.419]
NE	0.243	[0.143 0.351]	0.375	[0.192 0.612]	0.496	[0.389 0.614]	0.358	[0.288 0.446]
L2/L1 Norm	0.235	[0.108 0.362]	0.375	[0.192 0.557]	0.495	[0.392 0.627]	0.355	[0.293 0.417]
HM	0.240	[0.108 0.371]	0.375	[0.194 0.555]	0.494	[0.390 0.600]	0.357	[0.293 0.420]
PCA	0.181	[0.084 0.364]	0.483	[0.385 0.655]	0.695	[0.617 0.765]	0.426	[0.362 0.517]
KPCA	0.177	[0.054 0.360]	0.488	[0.375 0.668]	0.692	[0.576 0.773]	0.425	[0.375 0.512]
ISOMAP	0.142	[0.030 0.313]	0.456	[0.329 0.646]	0.673	[0.520 0.772]	0.395	[0.347 0.483]
CNN	0.139	[0.039 0.297]	0.247	[0.095 0.399]	0.362	[0.066 0.647]	0.239	[0.087 0.354]
GCV	0.145	[0.035 0.341]	0.397	[0.204 0.534]	0.644	[0.396 0.813]	0.370	[0.206 0.471]
GKLD	0.139	[0.031 0.328]	0.256	[0.045 0.535]	0.653	[0.534 0.834]	0.328	[0.199 0.458]
GWD	0.181	[0.103 0.384]	0.484	[0.351 0.620]	0.726	[0.593 0.858]	0.436	[0.333 0.491]
SIFE	0.169	[0.079 0.270]	0.326	[0.128 0.523]	0.468	[0.171 0.662]	0.306	[0.158 0.396]
SSN	0.240	[0.120 0.469]	0.648	[0.568 0.727]	0.788	[0.738 0.873]	0.527	[0.448 0.605]
RMSBT	0.243	[0.188 0.298]	0.589	[0.519 0.687]	0.775	[0.695 0.856]	0.507	[0.466 0.548]
Proposed	0.377	[0.214 0.553]	0.731	[0.659 0.795]	0.843	[0.791 0.891]	0.623	[0.520 0.693]

parameter of Gaussian distribution.

2.2.2. Distribution parameter alignment for EMM in frequency domain

The GMM can effectively characterize data distributions in time domain, however, it is less suitable for modeling frequency-domain data distributions. Therefore, the EMM is adopted to extract distributional information from frequency-domain data of WT bearing. The parameter corresponding to the exponential component with the maximum mixture weight λ is then extracted as $\lambda = \{\lambda_i\}_{i=1}^N$. To mitigate the effect of speed variations on this parameter, the distribution parameter

alignment is still required. The reference and non-reference values in λ are determined according to Eq. (5). The reference values of λ are then fitted using a third-degree polynomial function, expressed as:

$$f(\lambda) = a_1\lambda^3 + a_2\lambda^2 + a_3\lambda + b, \quad (10)$$

where a_1 , a_2 , a_3 , and b are model parameters. Using the fitted polynomial function, the reference values of λ at non-reference speeds can be estimated. Subsequently, the PAF parameters corresponding to the λ of exponential distribution are estimated based on the speed records, the

Table 2

Detailed configuration of different HI construction methods in ablation experiment.

Method	Description
GMM-DCRM	Estimating time-domain distribution parameters using GMM, constructing HI via DCRM
EMM-DCRM	Estimating frequency-domain distribution parameters using EMM, constructing HI via DCRM
GMM-EMM-DCRM	Estimating parameters in both domains via GMM and EMM, constructing time-domain and frequency-domain HIs separately via DCRM, then linearly weighting them to construct HI
GMM-DPA	Estimating time-domain distribution parameters using GMM, aligning parameters via DPA, constructing HI via DCRM
EMM-DPA	Estimating frequency-domain distribution parameters using EMM, aligning parameters via DPA, constructing HI via DCRM

raw parameter values, and the estimated reference values at non-reference speeds. Finally, the parameter λ after parameter alignment $\hat{\lambda} = \{\hat{\lambda}_i\}_{i=1}^N$ is obtained through the fitted PAF as follows:

$$\hat{\lambda}_i = \frac{\lambda_i - \beta_\lambda (\ln(s_i/s_R))}{(s_i/s_R)^{\alpha_\lambda}}, \quad (11)$$

where $\hat{\lambda}_i$ denotes the parameter of the exponential distribution after parameter alignment, α_λ and β_λ represent the estimated parameters of PAF corresponding to this parameter of exponential distribution.

2.3. Constructing WT bearing HI for degradation assessment

Using the acquired mean $\hat{\mu} = \{\hat{\mu}_i\}_{i=1}^N$ and variance $\hat{\sigma}^2 = \{\hat{\sigma}_i^2\}_{i=1}^N$, the data distribution after parameter alignment $\hat{\mathbf{p}}^T = \{\hat{p}_i^T\}_{i=1}^N$ for the monitoring data $\mathbf{y} = \{y_i\}_{i=1}^N$ can be acquired for HI construction. The health data located within the reference speed band in the first 50 observations are utilized to estimate the baseline distribution $\hat{p}_B^T$. Afterward, the distribution contact ratio metric (DCRM) is introduced to evaluate the distributional difference between $\hat{p}_B^T$ and $\hat{p}_i^T$, expressed as:

$$\hat{H}_i^T = \text{DCRM}(\hat{p}_B^T, \hat{p}_i^T) = 1 - \hat{O}_i^T, \quad (12)$$

where $\hat{H}_i^T$ denotes the DCRM-based HI value, and $\hat{O}_i^T$ represents the overlapping area between $\hat{p}_B^T$ and $\hat{p}_i^T$. The overlapping area $\hat{O}_i^T$ can be computed using the following expression:

$$\hat{O}_i^T = \int \min[\hat{p}_B^T(x), \hat{p}_i^T(x)] dx. \quad (13)$$

Finally, the THI vector $\hat{\mathbf{H}}^T = \{\hat{H}_i^T\}_{i=1}^N$ is constructed via the acquired DCRM values.

The THI can reflect the degradation evolution of WT bearing by analyzing time-domain distribution discrepancies. However, it is insufficient to comprehensively represent the degradation behavior when the deteriorative characteristics embedded in frequency-domain data are neglected. Therefore, FHI should be constructed to exploit the deterioration-related features contained in monitoring data. Based on the parameter λ after parameter alignment $\hat{\lambda} = \{\hat{\lambda}_i\}_{i=1}^N$, the data distributions after parameter alignment $\hat{\mathbf{p}}^F = \{\hat{p}_i^F\}_{i=1}^N$ for frequency-domain data are obtained for frequency-domain HI construction. The health frequency data located within the reference speed band in the first 50 observations are utilized to acquire the baseline distribution $\hat{p}_B^F$. The distributional difference between $\hat{p}_B^F$ and $\hat{p}_i^F$ is measured using the following formula:

$$\hat{H}_i^F = \text{DCRM}(\hat{p}_B^F, \hat{p}_i^F) = 1 - \hat{O}_i^F, \quad (14)$$

where $\hat{H}_i^F$ denotes the DCRM-based frequency-domain HI value, $\hat{O}_i^F$ represents the overlapping area between $\hat{p}_B^F$ and $\hat{p}_i^F$. The overlapping area $\hat{O}_i^F$ is calculated as:

$$\hat{O}_i^F = \int \min[\hat{p}_B^F(x), \hat{p}_i^F(x)] dx. \quad (15)$$

Accordingly, the FHI vector $\hat{\mathbf{H}}^F = \{\hat{H}_i^F\}_{i=1}^N$ is acquired. Based on the HIs constructed in the two domains, the DPA-HI of WT bearing $\mathbf{H} = \{H_i\}_{i=1}^N$ is derived by linearly weighting the two individual HIs as:

$$H_i = 0.5 \times \hat{H}_i^T + 0.5 \times \hat{H}_i^F. \quad (16)$$

2.4. IFD of WT bearing

The constructed HIs can provide a comprehensive representation of the deterioration evolution in WT bearings. Furthermore, these HIs can be further utilized to detect incipient fault point (IFP) for early fault warning. The alarm threshold is determined using the 3σ criterion, based on which the IFP of WT bearing can be accurately identified. After

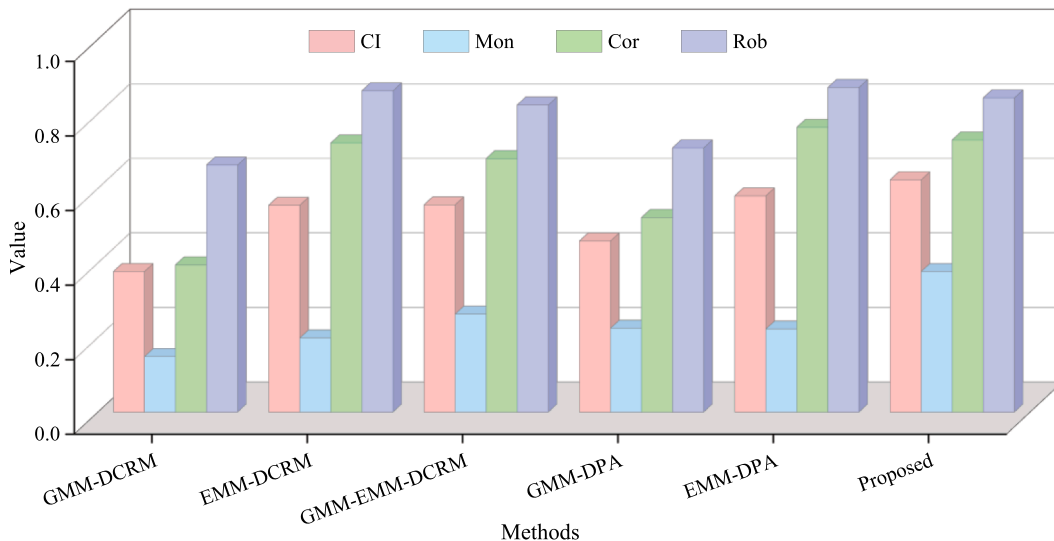


Fig. 15. The average evaluation results for all datasets.

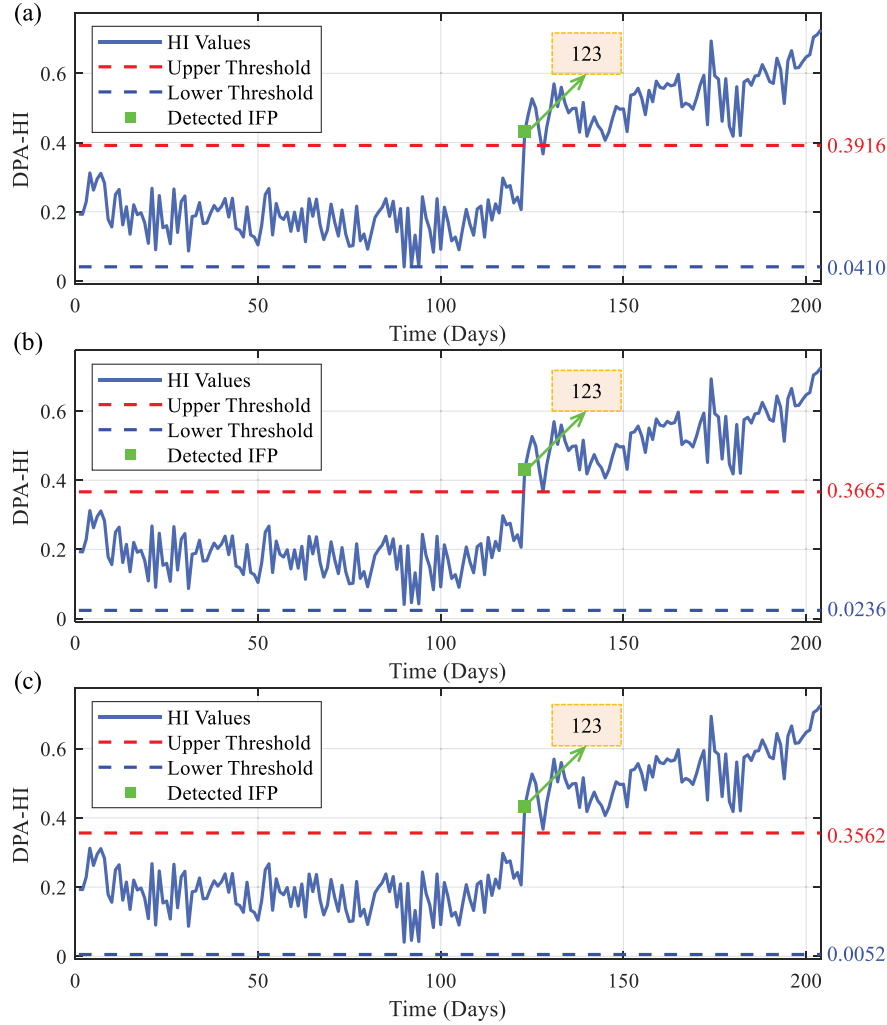


Fig. 16. The detection results for different baseline lengths. (a) 20 HI values, (b) 50 HI values, (c) 100 HI values.

identifying the IFP, the first data point among three consecutive observations exceeding the alarm threshold is defined as the true IFP. In contrast, isolated threshold exceedances that do not persist for at least three successive points are regarded as pseudo IFPs.

3. Experimental results and performance comparison

In this section, two WT bearing datasets are first introduced for validation purposes. Using these datasets, the HIs of WT bearings are constructed based on the proposed method. The effectiveness and superiority of the proposed DPA-HI are then validated through comparing with both conventional and advanced HIs. Finally, leveraging the constructed DPA-HI, IFD of WT bearings is achieved to enable early fault warning.

3.1. Dataset description

The two WT bearing datasets used for validation are described in detail in the following section, including WT generator bearing dataset and WT gearbox bearing dataset. Both datasets were collected from operational wind farms under real-world working conditions, providing a solid data foundation for evaluating the performance of the proposed DPA-HI.

3.1.1. WT generator bearing dataset

The WT generator bearing dataset was derived from a large-scale

wind farm located in the northern plains of China. The sensor configuration for data acquisition is illustrated in Fig. 2. As shown, accelerometers were mounted at both the driving end and non-driving end of the generator to acquire bearing vibration signals, which were recorded using the designed data acquisition system. The sampling frequency, sampling interval and acquisition duration were set to 25.6 kHz, 1 day, and 1.0 s, respectively. In this work, two WT generator bearing datasets (Generator 1 and Generator 2) are employed to validate the performance of the presented DPA-HI under TVSCs. After filtering out invalid samples, a total of 204 days of data from Generator 1 (collected from May 16 to November 3, 2018) and 225 days of data from Generator 2 (collected from May 15 to October 20, 2018) are applied in this experiment. The corresponding time-domain vibration signals and rotational speed profiles are presented in Figs. 3 and 4, respectively. Post-maintenance inspection revealed severe raceway wear and rolling element slippage, along with grease contamination caused by wear debris.

3.1.2. WT gearbox bearing dataset

The WT gearbox bearing dataset was acquired from a 2MW WT in a wind farm of China. The simplified schematic of this WT gearbox is displayed in Fig. 5. This gearbox comprises a main shaft, a planetary stage, and low-speed, intermediate, and high-speed shafts. The accelerometer was installed near the low-speed shaft to collect vibration responses. The vibration signals were sampled at 25.6 kHz with a recording length of 1.28 s for each sample, and the sampling interval was set to 1 day. Additionally, the rotational speed information was

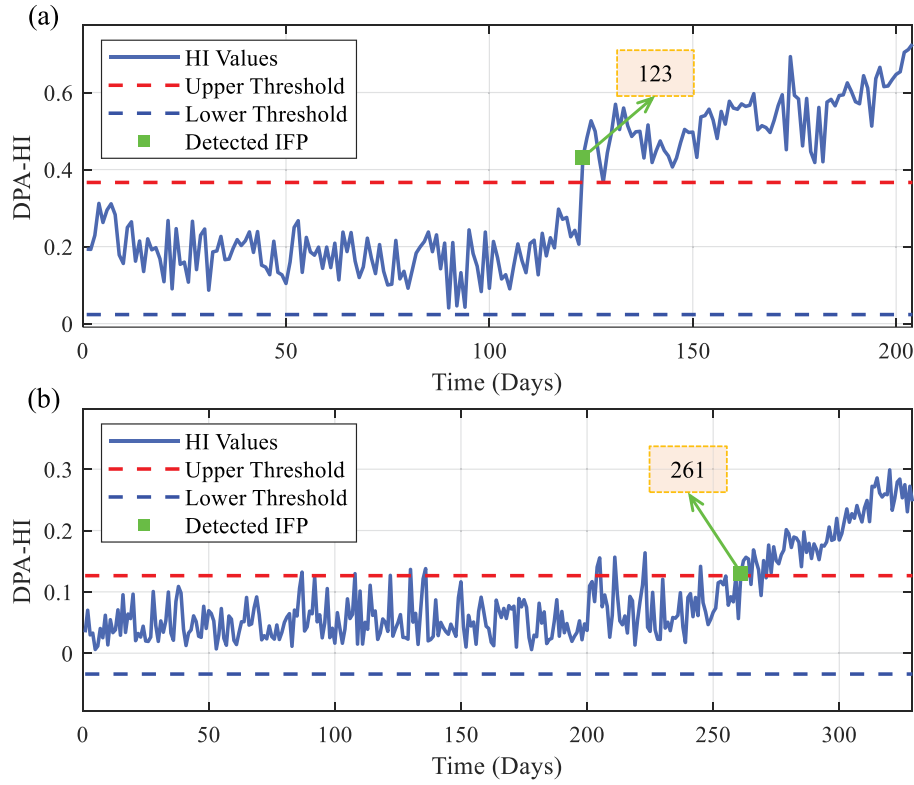


Fig. 17. The detection results for WT bearings. (a) Generator 1, (b) Gearbox 1.

Table 3
The FCFs of WT bearings.

Bearing types	Rotating (Hz)	BPFO (Hz)	BPFI (Hz)	BSF (Hz)	FTF (Hz)	Fault types discovered during maintenance
6334 M/C3	1	3.16	4.84	4.56	0.40	Raceway wear, rolling element slippage
23992CAMA/W33	1	17.70	20.30	7.23	0.47	Surface wear

collected via the rotational speed sensor. In this work, two WT gearbox bearing datasets (Gearbox 1 and Gearbox 2) are utilized to evaluate the performance and advantages of the developed DPA-HI. After discarding invalid samples, 329 days of data for Gearbox 1 (collected from January 1, 2018 to May 23, 2019) and 214 days of data for Gearbox 2 (collected from January 1, 2018 to May 8, 2019) are employed for experimental verification. The corresponding time-domain vibration signals are illustrated in Fig. 6, while the associated rotational speed profiles are shown in Fig. 7. During maintenance, serious wear faults were found on the bearing surfaces.

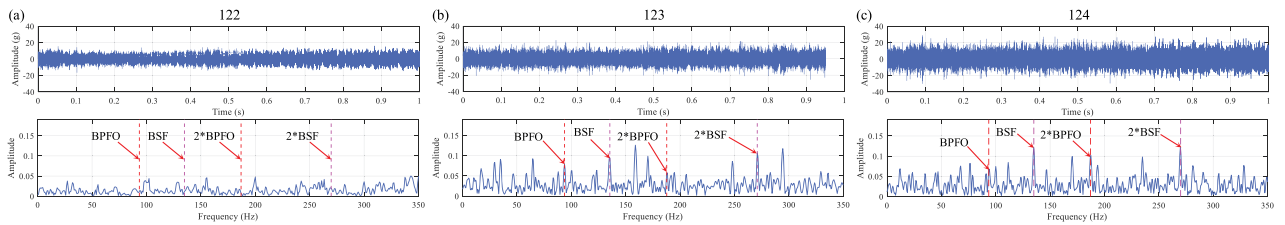


Fig. 18. The vibration signal waveforms and the corresponding spectra around position 123 for Generator 1.

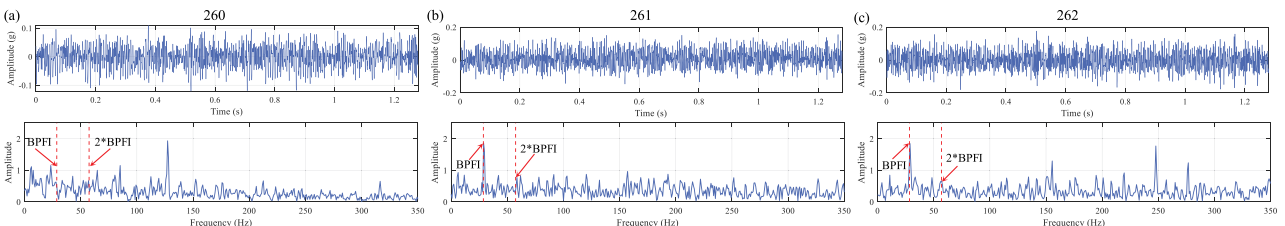


Fig. 19. The vibration signal waveforms and the corresponding spectra around position 261 for Gearbox 1.

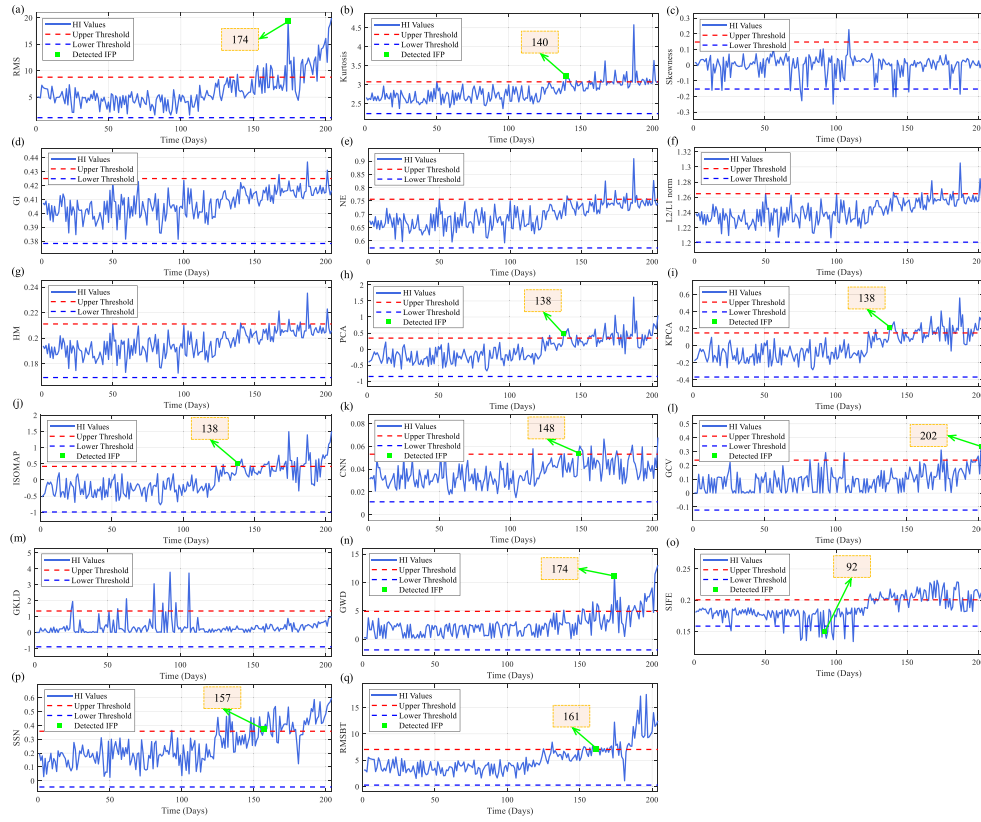


Fig. 20. The detection results for Generator 1. (a) RMS, (b) Kurtosis, (c) Skewness, (d) GI, (e) NE, (f) L2/L1 Norm, (g) HM, (h) PCA, (i) KPCA, (j) ISOMAP, (k) CNN, (l) GCV, (m) GKLD, (n) GWD, (o) SIFE, (p) SSN, (q) RMSBT.

Table 4

Quantitative evaluation of fault detection timeliness based on detection delay for Generator 1.

Method	RMS	Kurtosis	Skewness	GI	NE	L2/L1 norm	HM	PCA	KPCA
IFP	174	140	N.D.	N.D.	N.D.	N.D.	N.D.	138	138
Detection Delay	51	17	>81	>81	>81	>81	>81	15	15
Method	ISOMAP	CNN	GCV	GKLD	GWD	SIFE	SSN	RMSBT	Proposed
IFP	138	148	202	N.D.	174	92	157	161	123
Detection Delay	15	25	79	>81	51	−31	34	38	0

“N.D.” indicates that the method did not trigger any alarm within the entire life cycle (204 days).

3.2. HI construction

In this section, HIs of WT generator bearings are first constructed using the corresponding datasets. The constructed DPA-HI quantitatively characterizes the nonlinear performance degradation process of bearing under TVSCs. Moreover, it demonstrates superior performance compared to other HIs. To further evaluate its effectiveness and advantages, HIs of WT gearbox bearings are also constructed. The experimental results further validate the efficacy of the developed DPA-HI.

3.2.1. HI construction of WT generator bearing

The THIs of WT bearings are firstly constructed using the collected vibration signals. The GMM is configured with four components to estimate the data distributions of the monitoring signals. Based on the calculated reference speed and the defined reference speed band, the mean and variance parameters corresponding to non-reference speeds are aligned to the reference speed using the designed PAF. Consequently, the parameter-aligned data distributions are obtained for HI construction. The baseline distribution is then estimated using the health data within the reference speed band from the first 50 health samples and the established GMM. Subsequently, the THI is derived by measuring the distribution discrepancy using the DCRM.

For frequency-domain analysis, an EMM with five components is adopted to model the distributions of frequency-domain data. The exponential distribution parameter λ associated with the dominant component is extracted and subjected to parameter alignment. With the aid of the reference speed, the predefined speed band, and the developed PAF, the parameters at non-reference speeds are transformed to the reference speed condition, yielding aligned frequency-domain distributions for HI construction. Similarly, the baseline distribution is estimated using the health frequency-domain data within the reference speed band from the first 50 healthy samples. The FHI is subsequently obtained via the DCRM. Finally, the DPA-HI is derived by linearly fusing these two HIs. The acquired bearing HIs are illustrated in Fig. 8.

It can be observed from Fig. 8 that the acquired HI can quantitatively depict the nonlinear performance degradation process of generator bearing under TVSCs. During the normal operating stage, the HI remains at a low level, indicating that the current data distribution is highly consistent with the baseline distribution and that the bearing operates in a healthy state. As the operating time increases, the HI exhibits a gradual upward trend, demonstrating a growing discrepancy between the current and baseline distributions. This progressive divergence reflects the continuous degradation of the generator bearing, which eventually leads to functional failure.

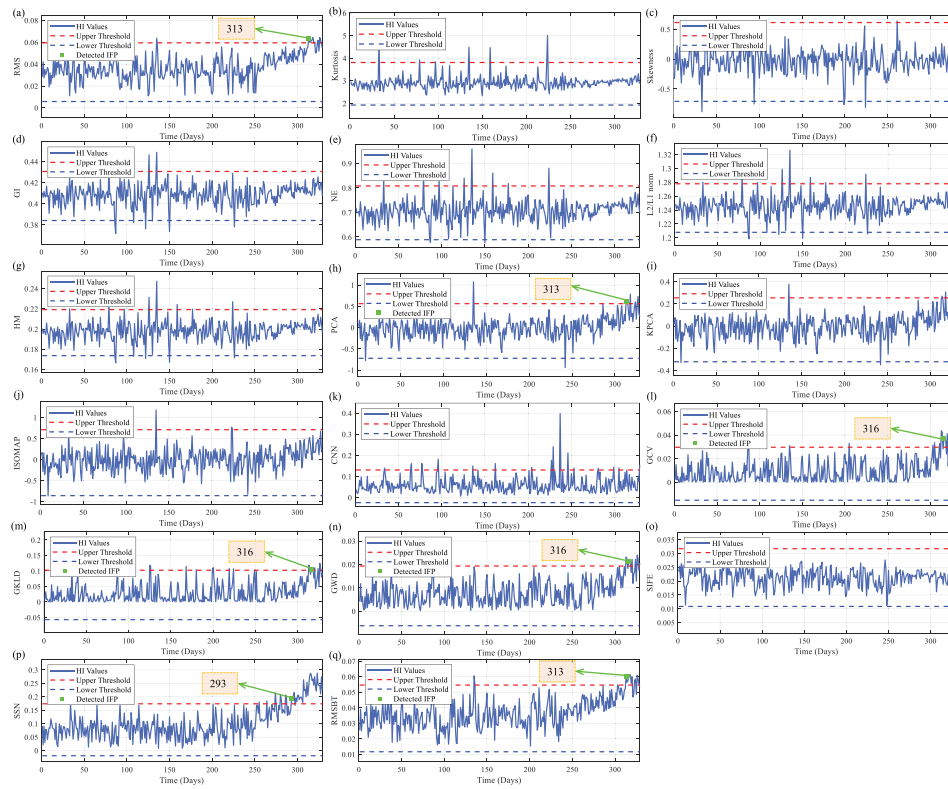


Fig. 21. The detection results for Gearbox 1. (a) RMS, (b) Kurtosis, (c) Skewness, (d) GI, (e) NE, (f) L2/L1 Norm, (g) HM, (h) PCA, (i) KPCA, (j) ISOMAP, (k) CNN, (l) GCV, (m) GKLD, (n) GWD, (o) SIFE, (p) SSN, (q) RMSBT.

Table 5

Quantitative evaluation of fault detection timeliness based on detection delay for Gearbox 1.

Method	RMS	Kurtosis	Skewness	GI	NE	L2/L1 norm	HM	PCA	KPCA
IFP	313	N.D.	N.D.	N.D.	N.D.	N.D.	N.D.	313	N.D.
Detection Delay	52	>68	>68	>68	>68	>68	>68	52	>68
Method	ISOMAP	CNN	GCV	GKLD	GWD	SIFE	SSN	RMSBT	Proposed
IFP	N.D.	N.D.	316	316	316	N.D.	293	313	261
Detection Delay	>68	>68	55	55	55	>68	32	52	0

“N.D.” indicates that the method did not trigger any alarm within the entire life cycle (329 days).

The conventional and advanced HI construction methods are employed for comparison. The conventional methods include RMS, kurtosis, skewness, GI [14], NE [15], L2/L1 Norm [16], HM [17], PCA [20], KPCA [21], and ISOMAP [22]. The advanced methods comprise CNN [25], GMM with confidence value (GCV) [41], GMM with Kullback-Leibler divergence (GKLD) [42], GMM with Wasserstein distance (GWD) [32], sigmoid informed fuzzy entropy (SIFE) [43], square speed normalization (SSN) [44], and RMS with baseline transformation (RMSBT) [38]. The performance of all HIs is quantitatively assessed using monotonicity (Mon), correlation (Cor), robustness (Rob), and the composite index (CI) [1]. Mon evaluates the consistency of the HI trend over time, where higher values indicate better monotonicity. Cor measures the linear correlation between the HI and time, with larger values denoting a stronger correlation. Rob assesses the tolerance of the HI to outliers, and higher values reflect better robustness. The CI provides an overall assessment of HI performance ($CI = 0.4 \times Mon + 0.3 \times Cor + 0.3 \times Rob$), where larger values signify superior overall performance. The evaluation results are displayed in Fig. 9. From Fig. 9(a), the presented DPA-HI achieves the maximum CI value of 0.605. Its Mon value of 0.241 surpasses all comparison methods, while the Cor and Rob values also exhibit superior performance. These results indicate that the presented DPA-HI can effectively characterize the overall deterioration trajectories of WT generator bearings under TVSCs. This observation

demonstrates that the distribution parameter alignment strategy successfully mitigates the effect of speed fluctuations on the distribution parameters, thereby enabling the constructed HI to accurately represent the intrinsic deterioration behavior of bearing. The similar conclusion is drawn from Fig. 9(b). Furthermore, the average evaluation results presented in Fig. 10 confirm that the proposed DPA-HI outperforms other HIs, providing reliable degradation assessment and facilitating accurate detection of incipient faults for WT bearings.

3.2.2. HI construction of WT gearbox bearing

To further evaluate the efficacy and merit of the developed HI construction approach, the WT gearbox bearing dataset is adopted to construct HI. The THI is derived by performing parameter alignment on the mean and variance parameters of the GMM, while the FHI is generated through alignment of the exponential parameter in the EMM. The DPA-HI is then generated by linearly fusing the two HIs. The generated HIs are shown in Fig. 11. As shown, the DPA-HI is capable of tracking the degradation trajectory of the gearbox bearing. In the health state, the HI values are relatively small, demonstrating that the bearing has not experienced any performance degradation. As degradation progresses, the HI values gradually increase, reflecting the onset and evolution of performance deterioration.

The comparative experiment is performed to assess the performance

of the presented HI in comparison with other HIs. The evaluation results are shown in Fig. 12. The results show that the developed HI achieves the highest values across all metrics, including CI, Mon, Cor, and Rob, in both cases. These findings demonstrate that the DPA-HI outperforms other HIs in overall performance and accurately reflects the dynamic degradation process under TVSCs. This confirms that the DPA method effectively mitigates the influence of speed fluctuations on the estimated distribution parameters. Furthermore, the average evaluation results presented in Fig. 13 indicate that the proposed HI exhibits superior monotonicity, temporal correlation, robustness, and comprehensive performance. Consequently, it can be reliably employed for precise detection of IFPs in WT gearbox bearings.

3.2.3. Performance analysis of dpa-hi for WT bearing

The constructed bearing HIs under non-reference speeds for Generator 1 are utilized to verify the effectiveness of the developed DPA method. To eliminate the influence of the differences in HI values before and after alignment, all HIs are normalized to the range [0, 1], ensuring they share the same data scale. The constructed HIs and rotational speed profiles under non-reference speeds are presented in Fig. 14. Specifically, Fig. 14(a) shows the THIs before and after alignment, Fig. 14(b) shows the FHIs before and after alignment, and Fig. 14(c) shows the final HIs before and after alignment. As observed in the figure, the trend fluctuations of the aligned HIs are smaller, whereas those of the unaligned HIs are more significantly affected by rotational speed. This indicates that the aligned HIs are less speed-dependent, demonstrating that the DPA method genuinely removes variable-speed effects from the HIs. By calculating the correlation between rotational speed and the final HIs, it can be known that the correlation coefficient before alignment is 0.41, while after alignment it is 0.17. This result further confirms that the DPA method substantially mitigates the influence of rotational speed, and that the established DPA-HI shows significantly reduced speed dependence. Nevertheless, some tolerable fluctuations remain in the healthy stage of the HI after DPA, which may attribute to inherent measurement noise and variations in the external operating environment.

The evaluation results and confidence intervals of different methods for all datasets are presented in Table 1 to quantitatively evaluate the performance of the proposed DPA-HI. The bootstrap confidence intervals are calculated by generating 5000 bootstrap samples, and the 95% confidence intervals are constructed using the bias corrected and accelerated percentile method. As shown in Table 1, the proposed method achieves the highest values across all four metrics (Mon=0.377, Cor=0.731, Rob=0.843, CI=0.623). Considering the confidence intervals, the lower bound of the proposed method for CI approaches or exceeds the upper bounds of most comparison methods, providing strong evidence of the performance improvement achieved by the proposed DPA-HI. It can be concluded that the developed DPA method effectively mitigates variable-speed effects in HI construction for WT bearings, thereby obtaining high-quality HIs for degradation assessment and IFD.

3.2.4. Ablation experiment

The HI construction methods, such as GMM with DCRM (GMM-DCRM), EMM with DCRM (EMM-DCRM), GMM and EMM with DCRM (GMM-EMM-DCRM), GMM with distribution parameter alignment (GMM-DPA), and EMM with distribution parameter alignment (EMM-DPA), are employed to conduct ablation experiment analysis. The detailed configuration of different methods are displayed in Table 2. The average evaluation results of all methods for all datasets are summarized in Fig. 15. As illustrated in Fig. 15, the CI of the presented HI is 0.623, which is higher than those of GMM-DCRM, EMM-DCRM, GMM-EMM-DCRM, GMM-DPA, and EMM-DPA. This indicates that the DPA-HI has superior overall performance. In addition, the DPA-HI outperforms the alternative HIs in terms of Mon, while maintaining Cor and Rob at levels comparable to the best-performing approaches. These results

demonstrate that the DPA-HI exhibits excellent monotonicity along with favorable temporal correlation and robustness. Moreover, the evaluation results of GMM-DPA and EMM-DPA are superior to those of GMM-DCRM and EMM-DCRM, respectively. This phenomenon confirms that the proposed DPA strategy can effectively mitigate the impact of speed variations, thereby enabling the generation of more reliable HIs for degradation assessment of WT bearings.

3.3. IFD of WT bearing

Before performing IFD of WT bearings, the baseline length must be determined because it directly affects the detection threshold. Taking Generator 1 as an example, a sensitivity analysis is conducted using three different baseline lengths (20, 50, and 100 HI values), with the corresponding detection results shown in Fig. 16. The detected IFP remains exactly the same across all three cases. Therefore, a baseline of 50 HI values is chosen to establish the 3σ interval $[\mu - 3\sigma, \mu + 3\sigma]$, as it balances statistical reliability with engineering practicality and adequately represents the healthy condition of the WT bearing.

The detection results are presented in Fig. 17. The detected IFPs of Generator 1 and Gearbox 1 are located at positions 123 and 261, respectively. The bearing types, fault characteristic frequencies (FCFs), and fault types discovered during maintenance are provided in Table 3, where the FCFs are computed at a generator shaft rotational speed of 1 Hz (60 RPM). For any given instantaneous shaft speed under variable-speed conditions, the corresponding instantaneous FCFs are obtained by $FCFs_{1Hz} \times \frac{s}{60}$, where $FCFs_{1Hz}$ are the FCFs shown in Table 3, s is the instantaneous shaft speed. To further validate these detections, the vibration signal waveforms and the corresponding spectra around the detected IFPs are displayed in Figs. 18 and 19, respectively. As observed in Fig. 18, there are indeed outer ring fault and rolling element fault at positions 123 and 124. From Fig. 19, an inner ring fault is evident at positions 261 and 262. Moreover, the FCFs is not prominent at positions 122 and 260, indicating that no faults occurred before the detected positions. These results demonstrate that the proposed DPA-HI can be efficiently applied to IFD of WT bearings, accurately identifying IFPs under TVSCs.

To further evaluate the detection performance of the developed DPA-HI in IFD, comparative HIs are adopted for IFP detection. The detection results for Generator 1 are presented in Fig. 20, and the quantitative evaluation of fault detection timeliness based on detection delay is summarized in Table 4. As observed, HIs constructed using skewness, GI, NE, L2/L1 norm, HM, and GKLD exhibit considerable fluctuations during the healthy operating stage, and these methods ultimately fail to detect any IFP within the entire lifecycle (denoted as "N.D." in Table 4). Such instability may lead to an enlarged threshold interval, potentially triggering false alarms. For methods that detect IFPs, including RMS, kurtosis, PCA, KPCA, ISOMAP, CNN, GCV, GWD, SSN, and RMSBT, their detection delays are all longer than that of the proposed DPA-HI, except for SIFE. The primary reason for these delays is that these methods do not adequately account for the effect of speed fluctuations on the distribution parameters of vibration signals. Although SIFE reports an earlier detection, the corresponding position does not correspond to an actual fault occurrence. In contrast, the proposed DPA-HI achieves the earliest true IFP detection with no delay.

Similarly, the detection results for Gearbox1 are illustrated in Fig. 21, and the quantitative evaluation of fault detection timeliness based on detection delay is summarized in Table 5. It is evident that HIs constructed using kurtosis, skewness, GI, NE, L2/L1 norm, HM, KPCA, ISOMAP, CNN, and SIFE exhibit significant amplitude variations under normal operating conditions, resulting in missed detections of the IFP within the entire lifecycle (denoted as "N.D." in Table 5). While RMS, PCA, GCV, GKLD, GWD, SSN, and RMSBT detected the IFP with positive detection delays ranging from 32 to 55 days. Such latency can compromise the timeliness of fault warnings during monitoring. In

contrast, the proposed DPA-HI detects the true IFP at position 261 with no delay. These observations highlight the effectiveness of the proposed DPA-HI in accurately and promptly identifying IFPs under TVSCs, outperforming both conventional and advanced comparative methods.

4. Conclusions

This paper developed a novel DPA-HI to achieve degradation assessment of WT bearings under TVSCs. Initially, the PAF was designed to align the distribution parameters from non-reference conditions to a predefined reference condition. Using the PAF, the non-reference distribution parameters of GMM and EMM were aligned to the reference speed condition. Subsequently, both THI and FHI were independently constructed using the aligned distribution parameters and DCRM. These two HIs are then linearly fused to derive the final DPA-HI for WT bearings. Finally, the efficacy and merit of the developed DPA-HI were verified using monitoring data from both WT generator and gearbox bearings. In addition, the constructed HI was also applied to IFD. Experimental results demonstrate that the proposed HI effectively captures the dynamic degradation trajectory of WT bearings. Moreover, its comprehensive performance outperforms both traditional and advanced HIs, facilitating accurate IFD under TVSCs. Nevertheless, the proposed method primarily focuses on mitigating the influence of speed variations on distribution parameters, while the effects of load variations are not explicitly considered. In addition, when rotational speed information is completely unavailable, the proposed method cannot be applied to construct HIs. Therefore, in future work, we will focus on jointly considering the effect of both speed and load on the distribution parameters, and developing methods that can alleviate the influence of time-varying operating conditions without relying on rotational speed, thereby enabling robust HI construction under broader practical scenarios.

CRedit authorship contribution statement

Dingliang Chen: Writing – original draft, Methodology, Funding acquisition. **Yan Peng:** Visualization, Validation, Funding acquisition, Formal analysis. **Jin Wang:** Writing – review & editing, Investigation, Conceptualization. **Yi Qin:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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